

01. Definition:

(i) Singular Value Decomposition :-

The singular value decomposition of a matrix A is the factorization of A into the product of three matrices $A = UDV^T$ where the column of U and V are orthogonal matrices and D is diagonal matrices with positive real entries.

(ii) Eigen Value Decomposition :-

In linear algebra, eigen decomposition or sometimes spectral decomposition is the factorization of a matrix, into a canonical form, whereby the matrix is represented in terms of its eigen values and eigen vectors. It can be written as, $A = PDP^{-1}$.

(iii) LU decomposition :-

If any linear system of equation can be performed as $AX = b$ without row interchanges, then A can be factored into the product of a lower triangular matrix L and an upper triangular matrix U . i.e. $A = LU$ which is known as LU decomposition or LU factorization.

(iv) Advantage of LU method :-

(1) More economic if we need to solve many systems with the same co-efficient matrix A but different right hand sides as we only need to evaluate L and U for one time. Once L and U are saved only the forward and backward substitution are needed to solve each system.

(2) Storage space may be economized. If A is not require after factorization, we can store L and U in A .

(v) Solve the linear system of using LU decomposition

$$\begin{aligned}u_1 + 2u_2 + 3u_3 &= 5 \\ 2u_1 - 4u_2 + 6u_3 &= 18 \\ 3u_1 - 9u_2 - 3u_3 &= 6\end{aligned}$$

Solution:-

Given that,

The linear system,

$$\begin{aligned}u_1 + 2u_2 + 3u_3 &= 5 \\ 2u_1 - 4u_2 + 6u_3 &= 18 \\ 3u_1 - 9u_2 - 3u_3 &= 6\end{aligned}$$

Which can be written as,

$$Au = b \dots \dots \dots (1)$$

Where,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}, \quad u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

Now, According to LU decomposition,

$$A = LU \dots\dots\dots (2)$$

Where,

$$L = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\therefore l_{11} = 1, \quad l_{11}u_{12} = 2 \quad \text{and } l_{11}u_{13} = 3$$

$$\Rightarrow u_{12} = 2 \quad \Rightarrow u_{13} = 3$$

Similarly,

$$l_{21} = 2, \quad l_{21}u_{12} + l_{22} = -4 \quad \text{and } l_{21}u_{13} + l_{22}u_{23} = 6$$

$$\Rightarrow l_{22} = -8 \quad \Rightarrow u_{23} = 0$$

$$l_{31} = 3, \quad l_{31}u_{12} + l_{32} = -9 \quad \text{and } l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3$$

$$\Rightarrow 3 \times 2 + l_{32} = -9 \quad \Rightarrow 3 \times 3 + (-15) \times 0 + l_{33} = -3$$

$$\Rightarrow l_{32} = -15 \quad \Rightarrow l_{33} = -12$$

Therefore,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

From (1) & (2)

We get, $LUu = b$ (3)

Let, $Uu = v$ (4)

Where, $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

Now, from (3) and (4),

$$Lv = b$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$\Rightarrow v_1 = 5$$

$$\therefore 2v_1 - 8v_2 = 18$$

$$3v_1 - 15v_2 - 12v_3 = 6$$

$$\therefore v_1 = 5$$

And, $2v_1 - 8v_2 = 18$

$$\Rightarrow 2 \times 5 - 8v_2 = 18$$

$$\Rightarrow v_2 = -1$$

Again, $3v_1 - 15v_2 - 12v_3 = 6$

$$\Rightarrow 3 \times 5 - 15(-1) - 12v_3 = 6$$

$$\Rightarrow v_3 = 2$$

Now, from (4), $Uu = v$

$$\Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow u_1 + 2u_2 + 3u_3 = 5$$

$$u_2 = -1$$

$$u_3 = 2$$

$$\therefore u_3 = 2$$

$$\therefore u_2 = -1$$

$$\text{And } u_1 + 2u_2 + 3u_3 = 5$$

$$\Rightarrow u_1 + 2(-1) + 3 \times 2 = 5$$

$$\Rightarrow u_1 = 1$$

$$\therefore (u_1, u_2, u_3) = (1, -1, 2)$$

2. Definition :-

(i) Vector :-

A vector is a quantity having both magnitude and direction. Such as displacement, velocity, force and acceleration etc.

(ii) Scalar :- A scalar is a quantity having magnitude but no direction. Such as mass, length, time, temperature and any real number.

(iii) Unit vector :-

A unit vector is a vector having unit magnitude. If $\vec{A}$ is a vector with magnitude $A \neq 0$, then $\vec{A}/A$ is a unit vector having the same direction as $\vec{A}$.

$$\text{Unit vector } \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

(iv) Dot product or (v) Scalar product :-

The Dot or scalar product of two vectors $\vec{A}$ and $\vec{B}$ denoted by $\vec{A} \cdot \vec{B}$ (read $\vec{A}$ dot $\vec{B}$) is defined as the product of magnitude of $\vec{A}$ and $\vec{B}$ and the cosine of the angle θ between them.

$$\text{In symbols, } \vec{A} \cdot \vec{B} = AB \cos \theta, \quad 0 \leq \theta \leq \pi$$

(vi) Vector field :-

If to each point (x, y, z) of a region R in space; there corresponds a vector $\vec{v}(x, y, z)$, then $\vec{v}$ is called a vector function and we say that a vector field $\vec{v}$ has been defined in R .

(vii) Scalar field :-

If to each point (x, y, z) of a region R in space there corresponds a number or scalar $\phi(x, y, z)$, then ϕ is called a scalar function of position or scalar point function and we say that a scalar field ϕ has been defined in R .